ADVANCED COGNITIVE NEUROSCIENCE

FROM SINGLE SUBJECT TO GROUP LEVEL ANALYSIS

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PHOTODIODE

Two purposes;

1. Check stimulus duration
2. Adjust trigger timing to when it was actually presented

For reporting timing of stim

Part of processing

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Adjusting trig timing in code (ish)

05_epochs.py

```
[...]

from utils.photodiode import (
    adjust_timing_of_events_photodiode)

[...]

if __name__ == "__main__":

    [...]
    events = mne.find_events(raw, min_duration=EVENTS_MIN_DURATION)

    if subject == "0188": # photodiode fell off
        adjusted_events[:, 0] = adjusted_events[:, 0] + 36
    else:
        adjusted_events = adjust_timing_of_events_photodiode(
            raw, events, EVENT_ID,
            report=report, photodiode_ch_name=PHOTODIODE_CH_NAME,
            stim_trigs=[1, 3]
        )
    [...]
    epochs = mne.Epochs(
        raw,
        events = adjusted_events
        [...]
        )
```


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MAXWELL FILTERING

What is it? Software for processing MEG data optimised for Elektra neuromag systems. Does a lot of things, for example:

- Signal space separation (SSS) or temporal signal space separation (tSSS)
  - “Software shielding”
- Compensating for head movements
- You (need) two files: which describe the calibration of the sensors and cross-talk

Reduces the rank of the data

- May reduce decoding accuracy

Want to know more?

- Videos introducing maxfilter
- Tutorial

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PRACTICALITIES

1. How far was the participant seated from the projector? **75 cm**
2. Participant demographics. **See overview.md on GitHub**
3. Explanation of PAS scores?

Table 1 The Perceptual Awareness Scale (PAS)

Label	Description [from Ramsøy and Overgaard (2004)]
(1) No Experience (NE)	No impression of the stimulus. All answers are seen as mere guesses.
(2) Weak Glimpse (WG)	A feeling that something has been shown. Not characterized by any content, and this cannot be specified any further.
(3) Almost Clear Experience (ACE)	Ambiguous experience of the stimulus. Some stimulus aspects are experienced more vividly than others. A feeling of almost being certain about one's answer.
(4) Clear Experience (CE)	Non-ambiguous experience of the stimulus. No doubt in one's answer.

Note: Scale steps and their descriptions.

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OCCIPITAL MEG ACTIVITY IN THE EARLY TIME RANGE (<300 MS) PREDICTS GRADED CHANGES IN PERCEPTUAL CONSCIOUSNESS

- Decoding PAS scores from *n* features (*n* = sources x time points) using logistic regression (L1 regularisation)
  - Accuracy per PAS score pair, brain area, time window (early/late) and participant.
- *"We fitted models with accuracy of the classifier as the dependent variable. Participants were modeled as having a unique intercept, that is, a random effect. Three fixed effects were of interest: PAS comparison (3 levels: NE–WG, WG–ACE, and ACE–CE), Lobe (5 levels: all, frontal, occipital, parietal, and temporal), and Time Range (2 levels: VAN and P3a; Fig. 3).*

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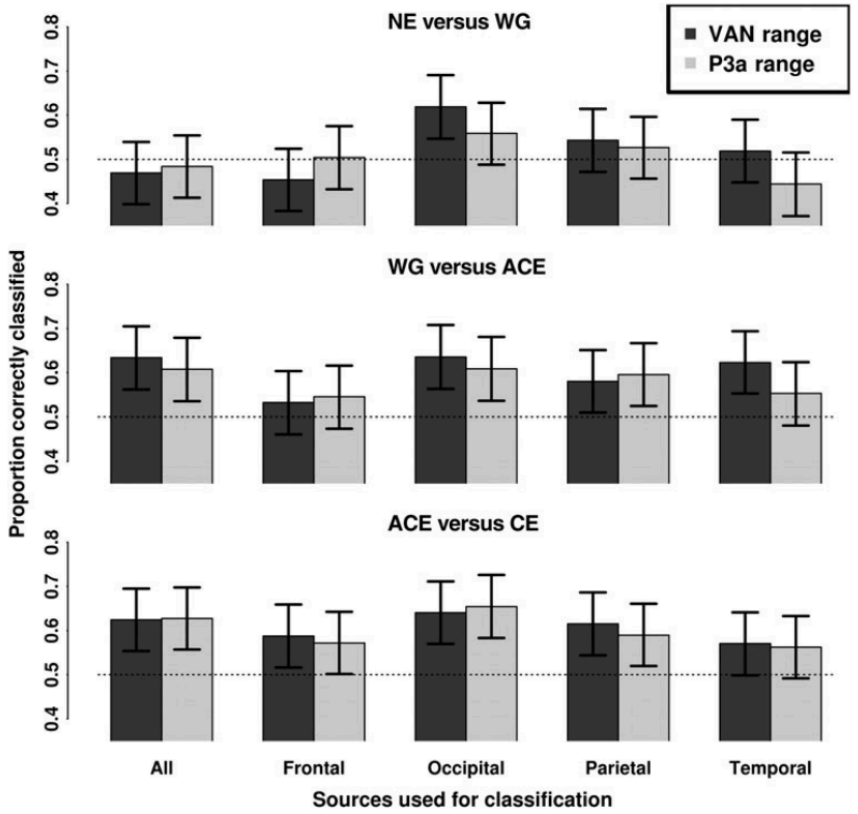


Figure 3. Mean classification accuracies for each of the 5 lobes tested for the 3 PAS comparisons for each of the 2 ranges: The VAN range (132–312 ms) and the P3a range (324–512 ms). NE versus WG is the difference of a subjective experience as such. WG versus ACE is the experiential difference of content. ACE versus CE is the experiential difference of unambiguity. Of special importance is it that occipital sources can be used to classify all PAS comparisons significantly above chance. The error bars are 95% confidence intervals tested against chance, bootstrapped using 10 000 simulations, from a mixed model having Time Range (2) and PAS comparison (3), and Lobe (5) as fixed effects including all possible interactions. Participants (11) were modeled with individual intercepts (random effect).

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- Additional ideas
  - Determining regularization strength using cross-validation (See `LogisticRegressionCV`)
  - Follow-up with time-resolved decoding to inform about time course

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MASTER SCRIPT

- Running scripts from a master script

```bash
#!/usr/bin/env bash
# Usage:
#   bash master.sh          # run all subjects
#   bash master.sh 0188     # run one subject

subjects=("0188" "0189" "0190" "0191" "0192" "0193" "0194" "0195")
run_group_stats=true

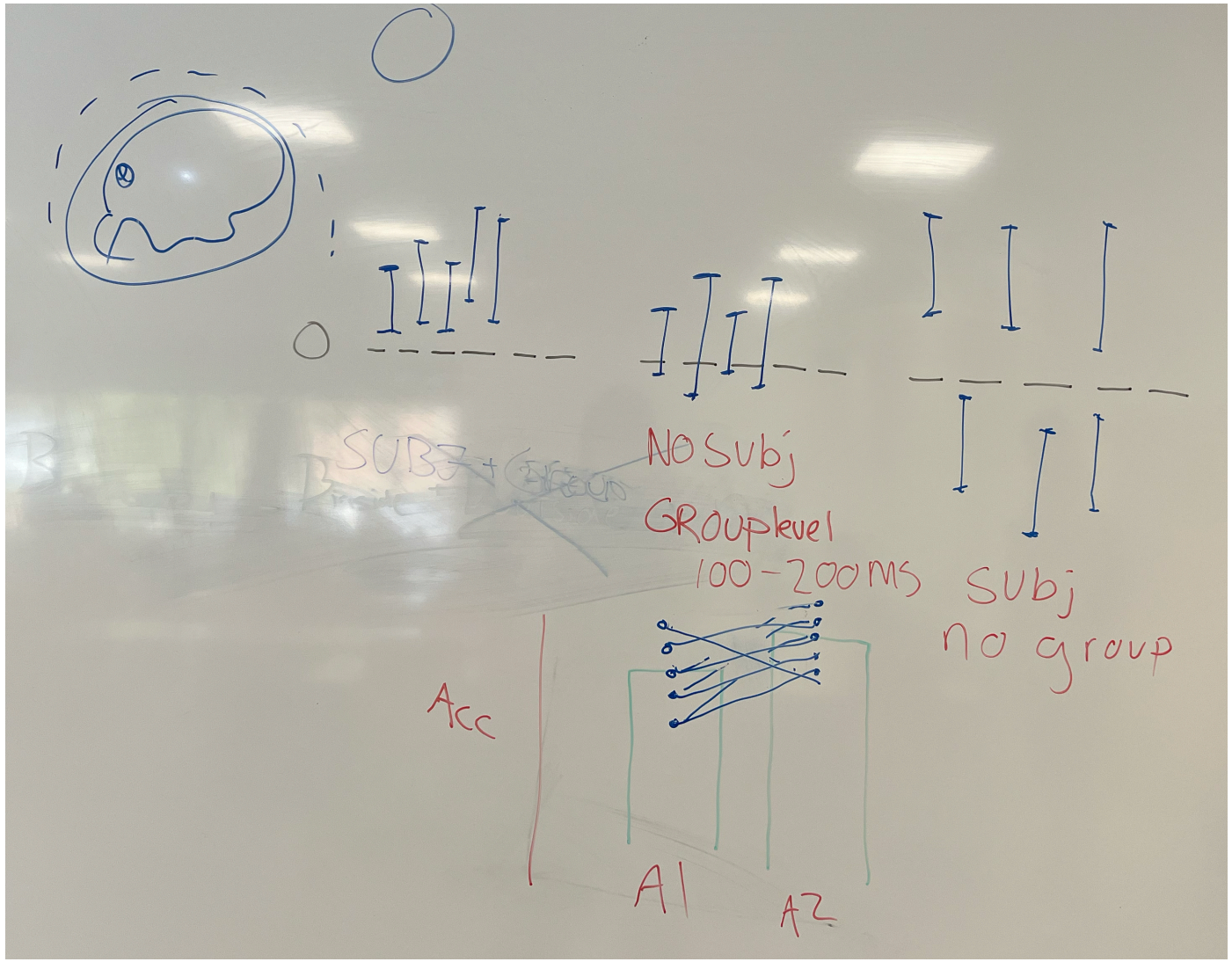
# If a subject is provided, override the list
if [ "$#" -eq 1 ]; then
subjects=("$1")

# Path to this script
SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"


for subject in "${subjects[@]}"; do
    echo "Running pipeline for subject: ${subject}"
    python "${SCRIPT_DIR}/01_anatomy.py" --subject "${subject}"
    python "${SCRIPT_DIR}/02_forward.py" --subject "${subject}"
    python "${SCRIPT_DIR}/03_filter.py"  --subject "${subject}"
    python "${SCRIPT_DIR}/04_ica.py"     --subject "${subject}"
    python "${SCRIPT_DIR}/05_epochs.py"  --subject "${subject}"
    python "${SCRIPT_DIR}/06_stcs.py"    --subject "${subject}"
    python "${SCRIPT_DIR}/07_evoked.py"  --subject "${subject}"

done

echo "Running group-level analysis"
python "${SCRIPT_DIR}/08_erf_stats.py"
```



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